

Keelan O'Doherty

Ann Arbor, MI | 602.376.2533 | keelan.odoherty@icloud.com | linkedin.com/in/keelanodoherty | github.com/Keelan4604 | U.S. Citizen, ITAR Eligible

EDUCATION

University of Michigan <i>Master of Science in Engineering (MSE), Aerospace Engineering, Space Systems</i> Relevant Coursework: Spacecraft Technology, Space Policy & Management, Linear Systems Theory, Management of Space Systems Design, Astrodynamics	Ann Arbor, MI · Aug 2026 – Dec 2027
University of Alabama, Honors College <i>Bachelor of Science (BS) in Aerospace Engineering, Minor in Mathematics</i> - Honors GPA 3.64/4.0 · Dean's List: every semester, Fall 2022 – Spring 2025 - Scholarships: Full-Ride Academic, Presidential, Engineering, Taylor Lewan Offensive Line Scholarship	Tuscaloosa, AL · Aug 2022 – May 2026
Florence University of the Arts, Study Abroad	Florence, Italy · May 2024

ENGINEERING EXPERIENCE

Northrop Grumman Sponsored Lunar Cargo Module (CAT-LIT) <i>Systems Engineer, SPACE 582 Spacecraft Design (Northrop Grumman sponsored)</i> - Drafted 48 Level 0 and Level 1 requirements for a standardized lunar cargo module (8,000 kg cargo, 2,000 kg dry, 5/7/8 m variants) compatible with Starship and Blue Moon, each traced to a parent with a verification method and success criteria (NASA SP-2016-6105) - Set the mission statement and six key performance parameters with threshold and objective values for the December mission design review and system requirements review - Authored the cargo history and interface standards section of the lunar cargo delivery survey; checked lander and launch vehicle envelope and load data against the Falcon, Vulcan, Ariane 6, New Glenn and Starship user guides	Ann Arbor, MI · Sep 2026 – Present
University of Alabama Aerodynamics Research Laboratory (AAERO) <i>Undergraduate Research Assistant, Vertical Flight Society Chapter</i> - Designed, SLA 3D-printed, and bench-tested 4 UAV propeller geometries at 10,000 RPM to characterize thrust against acoustic performance - Built the test setup and authored run procedures using load-cell thrust instrumentation and microphone acoustic profiling, structured for repeatable comparison across geometries - Correlated measured results against Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) predictions to identify model-to-hardware divergence; presented findings at the URCA Conference 2025	Tuscaloosa, AL · Dec 2024 – May 2026
Fixed-Wing eVTOL Aircraft - Design, Build & Flight Test - Modeled the full ORCA airframe in SolidWorks from scratch for 3D printing, adapted from an existing VTOL design (fuselage, wings, spars, control surfaces, motor mounts), including a modular payload nose for optical, IR, and thermal sensor swaps - Served as team pilot and ran the flight test campaign: phased hover and vertical-to-forward transition trials validating stability, control response, and endurance - Modeled propulsion tilt-path dynamics and servo actuation sequencing in MATLAB/Simulink; validated the power budget against measured draw	Senior Capstone · Aug 2025 – May 2026
Wilson & Company <i>Engineering Intern</i> Gained hands-on experience with construction materials and techniques through 50+ hours of on-site exposure, documenting nonconformances and as-built conditions	Phoenix, AZ · Jul 2024 – Aug 2024

ADDITIONAL EXPERIENCE

Watercraft Restoration - Owner & Technician - Restored 65+ watercraft end to end, including 30+ full engine rebuilds and 30+ complete electrical rewires, from teardown through functional test - Fabricated wire harnesses and mechanical subassemblies to torque specification; wrote repeatable teardown, build, and checkout procedures that reduced rework on later units	Phoenix, AZ · Feb 2021 – Aug 2026
Pi Kappa Alpha Fraternity - Class President; Sports Chair	Tuscaloosa, AL · 2022 – 2026

TECHNICAL PROJECTS

Embedded Vehicle Control System - built Arduino-based control with GSM/IoT telemetry (SIM7000A) over LTE CAT-M1 for remote monitoring and actuation; designed power distribution using transistor logic and relay control, and wired the physical harness
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TECHNICAL SKILLS

CAD: SolidWorks (90-part ORCA airframe assembly with wing, tail and fuselage subassemblies built for 3D printing; motor-mount and spar revisions; nozzle geometry exported to STEP for CFD), FDM 3D printing, wire harness design
Mission & Orbit Analysis: STK 12 (sun-synchronous and mid-inclination coverage scenarios with rectangular and conic sensors, revisit and N-asset figures of merit; Astrogator lunar transfer with differential-corrector TLI and B-plane targeted LOI; low-thrust spiral with a custom propagator), patched-conic and Hohmann analysis in MATLAB
CFD & Simulation: ANSYS Fluent (axisymmetric density-based nozzle flow, SST k-omega, four-level mesh refinement to 137k cells; NACA 2412 with Spalart-Allmaras and SST; heated-channel convection), MATLAB/Simulink (DC-8 state-space pitch and roll-yaw control loops, quasi-1D nozzle solver, intercept simulation, solid and liquid stage sizing)
Programming & Tools: Python (NumPy, SciPy, Jupyter: laminate and buckling analysis, design searches), MATLAB, Arduino/C++, Git/GitHub, Excel, Microsoft 365
Additional: Languages: Spanish · Musical: Piano, Cello, Guitar